

COVID-19 Data Analysis

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Introduction and Questions of Interest

This report analyzes the Johns Hopkins University CSSE COVID-19 dataset, which tracks confirmed cases and deaths from January 22, 2020 through March 9, 2023, when JHU stopped collecting data. The data comes as four time series files in wide format: one row per location (country or province globally; county for the US) and one column per date, holding *cumulative* counts. A separate lookup table supplies populations, which we join in to enable per-capita comparisons.

The analysis addresses three questions:

1. How did cases and deaths evolve over time in the US, and what distinct waves are visible?
2. How does the US compare to other countries, and how much do outcomes vary across US states, on a per-capita basis?
3. Does a state's case rate predict its death rate? ie. If we know how many infections a state had per capita, how well can we predict its deaths per capita?

Required libraries: tidyverse and lubridate.

```
library(tidyverse)
library(lubridate)
```

Importing the Data

The data comes from the Johns Hopkins CSSE COVID-19 repository on GitHub. We read in four time series files (global and US, cases and deaths) plus a lookup table containing population data. Reading directly from the URLs keeps the analysis fully reproducible.

```
url_in <- paste0("https://raw.githubusercontent.com/CSSEGISandData/",
                 "COVID-19/master/csse_covid_19_data/",
                 "csse_covid_19_time_series/")

file_names <- c("time_series_covid19_confirmed_global.csv",
                "time_series_covid19_deaths_global.csv",
                "time_series_covid19_confirmed_US.csv",
                "time_series_covid19_deaths_US.csv")

urls <- str_c(url_in, file_names)

global_cases <- read_csv(urls[1])
global_deaths <- read_csv(urls[2])
us_cases <- read_csv(urls[3])
us_deaths <- read_csv(urls[4])

uid_lookup_url <- paste0("https://raw.githubusercontent.com/CSSEGISandData/",
                        "COVID-19/master/csse_covid_19_data/",
```

```

                                "UID_ISO_FIPS_LookUp_Table.csv")
uid <- read_csv(uid_lookup_url) %>%
  select(-c(Lat, Long_, Combined_Key, code3, iso2, iso3, Admin2))

```

Tidying and Transforming the Data

The raw files are in wide format, with one column per date. We pivot each date column into rows, convert the dates from character to date objects, drop the latitude/longitude columns we will not use, and join cases with deaths. For the global data we also join in populations from the lookup table so per-capita comparisons are possible.

```

global_cases <- global_cases %>%
  pivot_longer(cols = -c(`Province/State`, `Country/Region`, Lat, Long),
               names_to = "date",
               values_to = "cases") %>%
  select(-c(Lat, Long))

global_deaths <- global_deaths %>%
  pivot_longer(cols = -c(`Province/State`, `Country/Region`, Lat, Long),
               names_to = "date",
               values_to = "deaths") %>%
  select(-c(Lat, Long))

global <- global_cases %>%
  full_join(global_deaths) %>%
  rename(Country_Region = `Country/Region`,
         Province_State = `Province/State`) %>%
  mutate(date = mdy(date)) %>%
  filter(cases > 0) %>%
  unite("Combined_Key",
        c(Province_State, Country_Region),
        sep = ", ",
        na.rm = TRUE,
        remove = FALSE) %>%
  left_join(uid, by = c("Province_State", "Country_Region")) %>%
  select(Province_State, Country_Region, date,
        cases, deaths, Population, Combined_Key)

summary(global)

```

```

##   Province_State   Country_Region      date      cases
## Length :306827   Length :306827   Min.   :2020-01-22   Min.    :      1
## N.unique :    90   N.unique :   201   1st Qu.:2020-12-12   1st Qu.:   1316
## N.blank  :     0   N.blank  :    0    Median :2021-09-16   Median :   20365
## Min.nchar:     4   Min.nchar:     2    Mean  :2021-09-11   Mean  :  1032863
## Max.nchar:    44   Max.nchar:    32   3rd Qu.:2022-06-15   3rd Qu.:  271281
## NAs      :210689           Max.   :2023-03-09   Max.   :103802702
##
##      deaths      Population      Combined_Key
## Min.   :      0   Min.   :6.700e+01   Length :306827
## 1st Qu.:      7   1st Qu.:7.866e+05   N.unique :   288
## Median :   214   Median :6.948e+06   N.blank  :     0
## Mean   :  14405   Mean   :2.890e+07   Min.nchar:     2
## 3rd Qu.:   3665   3rd Qu.:2.914e+07   Max.nchar:    60

```

```
## Max.      :1123836   Max.      :1.380e+09
##                               NAs      :6729
```

```
us_cases <- us_cases %>%
  pivot_longer(cols = -(UID:Combined_Key),
               names_to = "date",
               values_to = "cases") %>%
  select(Admin2:cases) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))
```

```
us_deaths <- us_deaths %>%
  pivot_longer(cols = -(UID:Population),
               names_to = "date",
               values_to = "deaths") %>%
  select(Admin2:deaths) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))
```

```
us <- us_cases %>%
  full_join(us_deaths)
```

```
summary(us)
```

```
##      Admin2      Province_State      Country_Region
## Length :3819906 Length :3819906 Length :3819906
## N.unique : 1980 N.unique : 58 N.unique : 1
## N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 3 Min.nchar: 4 Min.nchar: 2
## Max.nchar: 41 Max.nchar: 24 Max.nchar: 2
## NAs : 6858
##      Combined_Key      date      cases      Population
## Length :3819906 Min. :2020-01-22 Min. : -3073 Min. : 0
## N.unique : 3342 1st Qu.:2020-11-02 1st Qu.: 330 1st Qu.: 9917
## N.blank : 0 Median :2021-08-15 Median : 2272 Median : 24892
## Min.nchar: 8 Mean :2021-08-15 Mean : 14088 Mean : 99604
## Max.nchar: 55 3rd Qu.:2022-05-28 3rd Qu.: 8159 3rd Qu.: 64979
## Max. :2023-03-09 Max. :3710586 Max. :10039107
##      deaths
## Min. : -82.0
## 1st Qu.: 4.0
## Median : 37.0
## Mean : 186.9
## 3rd Qu.: 122.0
## Max. :35545.0
```

Next we aggregate the US county-level data up to the state level and to the national level, and calculate deaths per million for per-capita comparison.

```
us_by_state <- us %>%
  group_by(Province_State, Country_Region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(Province_State, Country_Region, date,
         cases, deaths, deaths_per_mill, Population) %>%
```

```

ungroup()

us_totals <- us_by_state %>%
  group_by(Country_Region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(Country_Region, date,
         cases, deaths, deaths_per_mill, Population) %>%
  ungroup()

tail(us_totals)

## # A tibble: 6 x 6
##   Country_Region date          cases  deaths deaths_per_mill Population
##   <chr>          <date>      <dbl>   <dbl>      <dbl>      <dbl>
## 1 US            2023-03-04 103650837 1122172      3371.    332875137
## 2 US            2023-03-05 103646975 1122134      3371.    332875137
## 3 US            2023-03-06 103655539 1122181      3371.    332875137
## 4 US            2023-03-07 103690910 1122516      3372.    332875137
## 5 US            2023-03-08 103755771 1123246      3374.    332875137
## 6 US            2023-03-09 103802702 1123836      3376.    332875137

```

By the end of the dataset (March 9, 2023), the US had recorded roughly 103.8 million cases and 1.12 million deaths, out of a global total of about 676.6 million cases and 6.88 million deaths.

Visualizing the Data

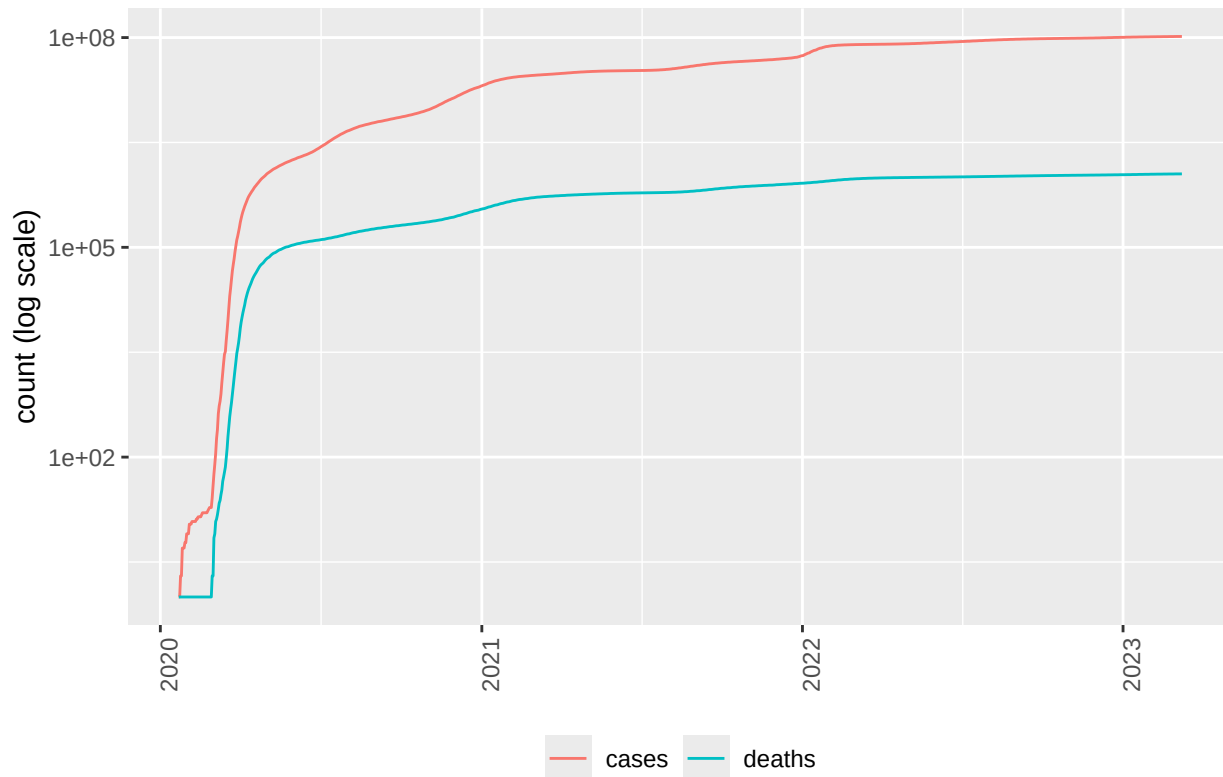
Cumulative cases and deaths in the US

```

us_totals %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID-19 in the US: Cumulative Cases and Deaths",
       y = "count (log scale)", x = NULL, color = NULL)

```

COVID-19 in the US: Cumulative Cases and Deaths



On the log scale, both curves rise rapidly in US Spring 2020 and then flatten. Cumulative counts, however, hide the wave structure of the pandemic, so we look at *new* cases instead. The coverage of why the wave-structure of the pandemic is important has an example here: [https://www.cell.com/heliyon/fulltext/S2405-8440\(24\)01121-6?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2405844024011216%3Fshowall%3Dtrue](https://www.cell.com/heliyon/fulltext/S2405-8440(24)01121-6?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2405844024011216%3Fshowall%3Dtrue). The wave structure indicates both the evolution and expected progression of the virus.

New cases: the wave structure of the pandemic

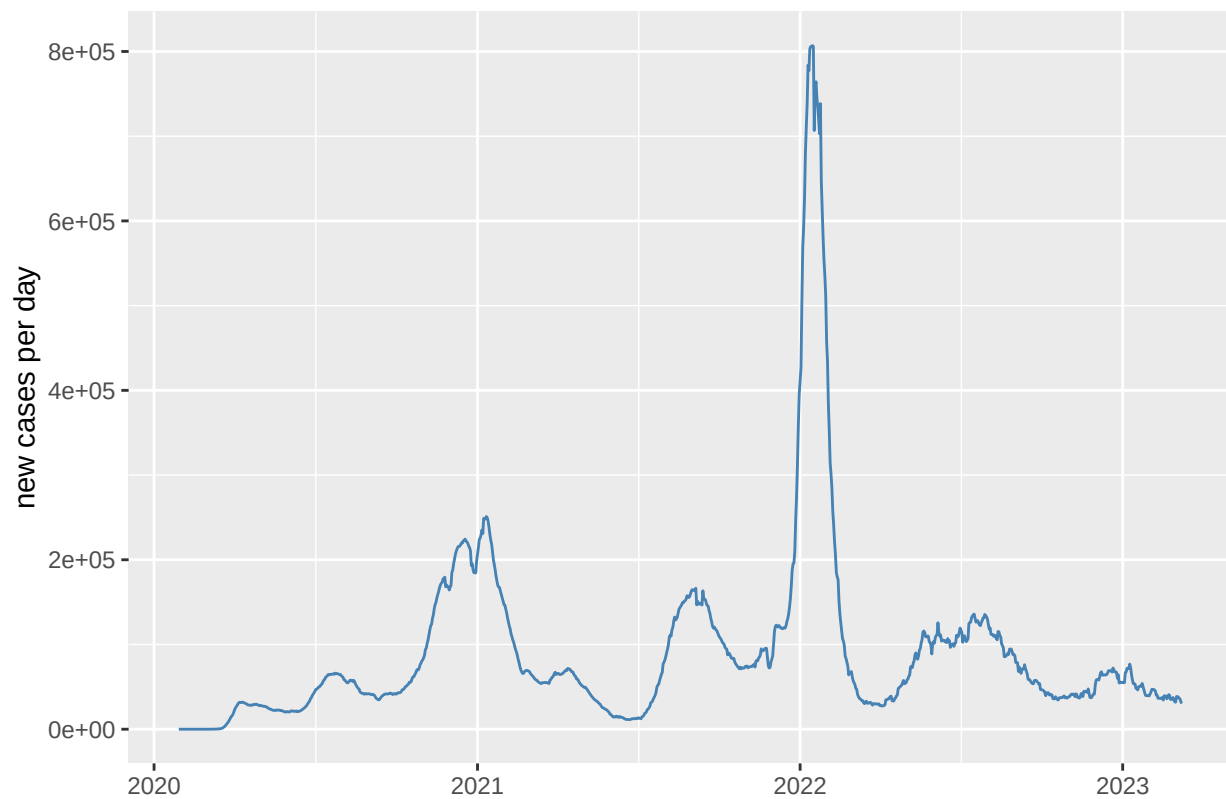
Because cases and deaths are cumulative, we compute daily new counts with `lag()`, then smooth with a 7-day rolling average to remove the strong day-of-week reporting cycle (a bias listed below).

```
us_totals <- us_totals %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths))

us_totals <- us_totals %>%
  mutate(new_cases_7day = stats::filter(new_cases, rep(1/7, 7), sides = 1),
         new_deaths_7day = stats::filter(new_deaths, rep(1/7, 7), sides = 1))

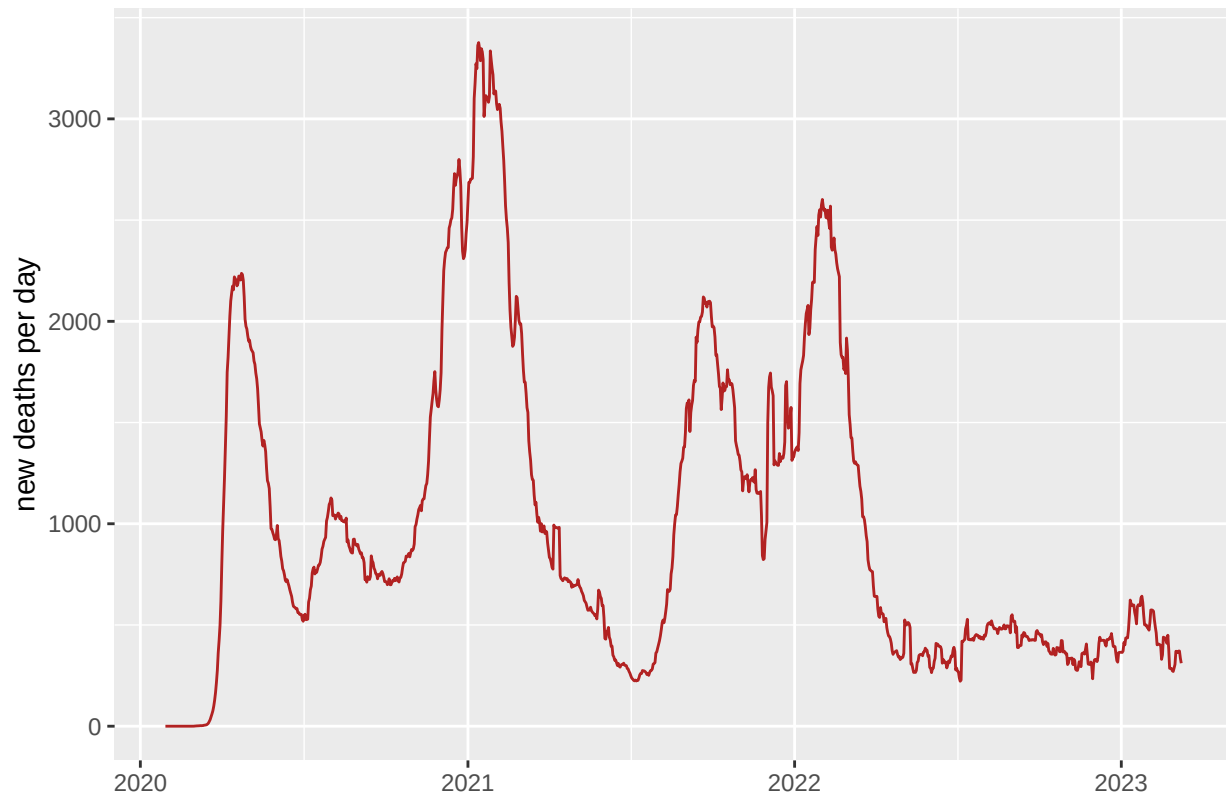
us_totals %>%
  filter(!is.na(new_cases_7day)) %>%
  ggplot(aes(x = date, y = as.numeric(new_cases_7day))) +
  geom_line(color = "steelblue") +
  theme(legend.position = "bottom") +
  labs(title = "US Daily New Cases (7-day average)",
       y = "new cases per day", x = NULL)
```

US Daily New Cases (7-day average)



```
us_totals %>%  
  filter(!is.na(new_deaths_7day)) %>%  
  ggplot(aes(x = date, y = as.numeric(new_deaths_7day))) +  
  geom_line(color = "firebrick") +  
  labs(title = "US Daily New Deaths (7-day average)",  
        y = "new deaths per day", x = NULL)
```

US Daily New Deaths (7-day average)



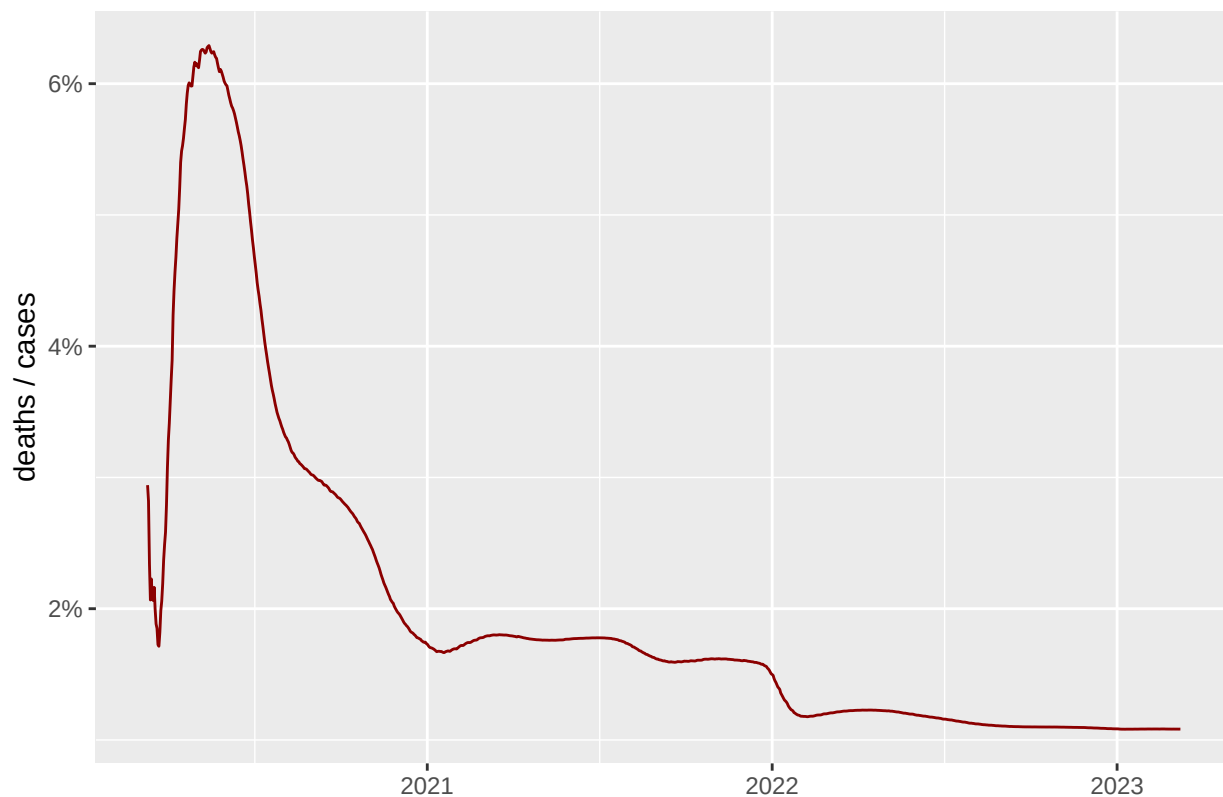
Several distinct waves are visible: spring 2020, summer 2020, the large winter 2020-21 wave, the Delta wave in late summer 2021, and the enormous Omicron spike, which peaked at roughly 807,000 new cases per day in mid-January 2022. Notably, the deaths curve does **not** mirror the cases curve: daily deaths peaked at about 3,400 per day in January 2021, a year *before* the case peak. The Omicron wave produced far more cases but proportionally fewer deaths, consistent with vaccination, prior immunity, and a less lethal variant.

Case fatality ratio over time

We can separate the wave-structure and fatality by looking at the cumulative case fatality ratio (CFR), deaths divided by cases. See: <https://www.britannica.com/science/case-fatality-rate>

```
us_totals %>%
  filter(cases > 1000) %>%
  mutate(cfr = deaths / cases) %>%
  ggplot(aes(x = date, y = cfr)) +
  geom_line(color = "darkred") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "US Cumulative Case Fatality Ratio",
       y = "deaths / cases", x = NULL)
```

US Cumulative Case Fatality Ratio



The cumulative CFR fell from about 1.7% at the end of 2020 to about 1.1% by the end of the dataset. The early figure is inflated by limited testing (many mild cases were never counted), and the later decline reflects vaccines, better treatment, and the milder Omicron variant.

How does the US compare globally?

Using the tidied global data (with populations joined in), we compute each country's cumulative deaths per thousand residents and compare the US against a set of peer countries, restricting to countries with over one million people so small-population outliers don't dominate.

```
country_totals <- global %>%
  group_by(Country_Region) %>%
  filter(date == max(date)) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population, na.rm = TRUE)) %>%
  filter(Population > 1000000) %>%
  mutate(deaths_per_thou = 1000 * deaths / Population,
         cases_per_thou = 1000 * cases / Population)

comparison_countries <- c("US", "United Kingdom", "Italy", "France",
                          "Germany", "Canada", "Japan", "Korea, South",
                          "Mexico", "Brazil", "India", "Peru",
                          "Australia", "Sweden")

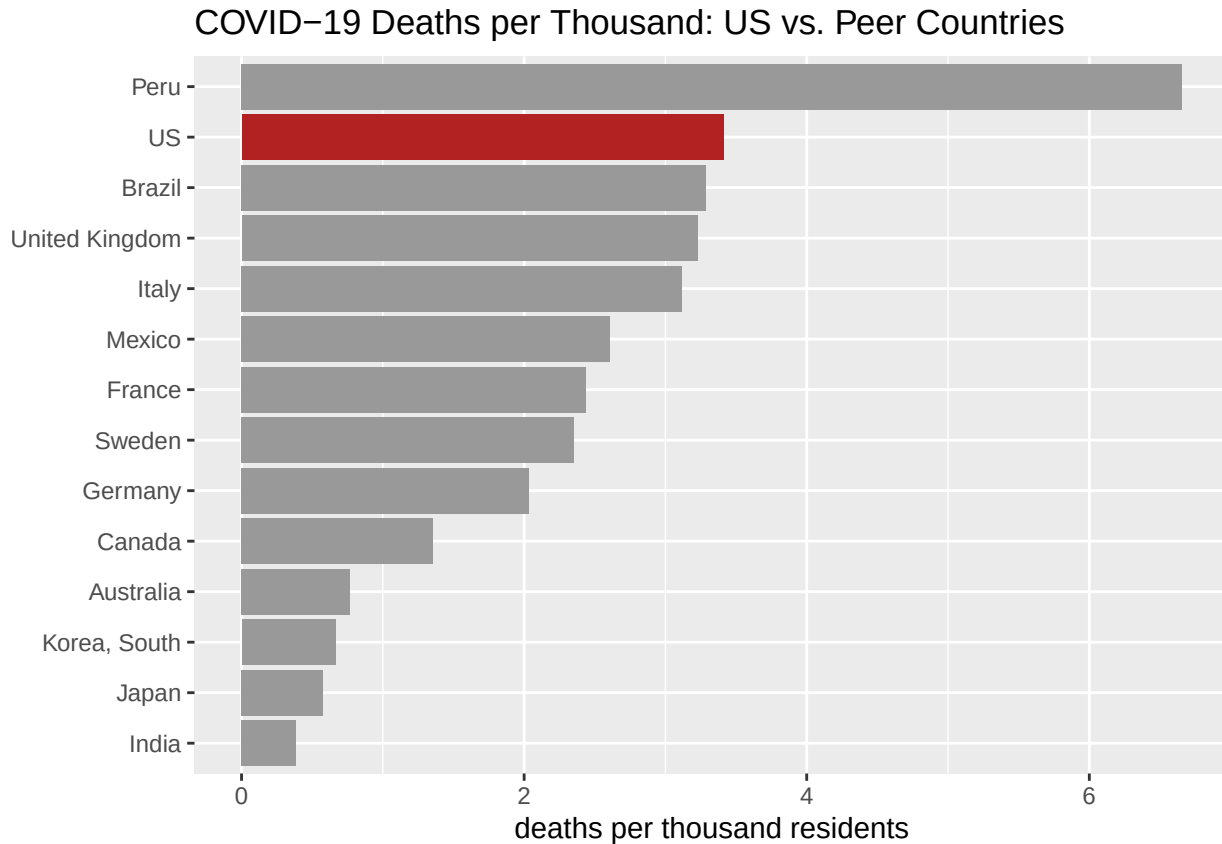
country_totals %>%
  filter(Country_Region %in% comparison_countries) %>%
  ggplot(aes(x = reorder(Country_Region, deaths_per_thou),
              y = deaths_per_thou,
```



```

    fill = Country_Region == "US")) +
  geom_col(show.legend = FALSE) +
  scale_fill_manual(values = c("grey60", "firebrick")) +
  coord_flip() +
  labs(title = "COVID-19 Deaths per Thousand: US vs. Peer Countries",
       x = NULL, y = "deaths per thousand residents")

```



```

country_totals %>%
  slice_max(deaths_per_thou, n = 10) %>%
  select(Country_Region, deaths_per_thou, cases_per_thou)

```

```

## # A tibble: 10 x 3
##   Country_Region      deaths_per_thou cases_per_thou
##   <chr>              <dbl>         <dbl>
## 1 Peru                6.66           136.
## 2 Bulgaria            5.50           187.
## 3 Hungary             5.05           227.
## 4 Bosnia and Herzegovina 4.96           122.
## 5 North Macedonia     4.64           166.
## 6 Croatia             4.38           309.
## 7 Georgia             4.25           458.
## 8 Czechia             3.97           431.
## 9 Slovakia            3.87           491.
## 10 Lithuania          3.52           480.

```

At about 3.4 deaths per thousand, the US fared worse than nearly every wealthy peer — the UK, Italy, and France sit near or below 3.2, Germany at 2.0, Canada at 1.4, and Japan and South Korea below 0.7 — and ranks roughly 12th worst among the 157 countries with over a million people, in the company of Peru (6.7,

the world's worst) and the hardest-hit countries of Eastern Europe. This comparison must be read cautiously: reported deaths dramatically undercount true mortality in countries with weaker reporting fidelity, so the honest conclusion is that the US did poorly *relative to countries with comparable reporting systems*, not that it did worse than most of the world.

Which states fared best and worst?

```
us_state_totals <- us_by_state %>%
  group_by(Province_State) %>%
  summarize(deaths = max(deaths), cases = max(cases),
            population = max(Population),
            cases_per_thou = 1000 * cases / population,
            deaths_per_thou = 1000 * deaths / population) %>%
  filter(cases > 0, population > 0)

us_state_totals %>%
  slice_max(deaths_per_thou, n = 10) %>%
  select(Province_State, deaths_per_thou, cases_per_thou)
```

```
## # A tibble: 10 x 3
##   Province_State deaths_per_thou cases_per_thou
##   <chr>          <dbl>          <dbl>
## 1 Arizona        4.55          336.
## 2 Oklahoma        4.54          326.
## 3 Mississippi    4.49          333.
## 4 West Virginia  4.44          359.
## 5 New Mexico     4.32          320.
## 6 Arkansas        4.31          334.
## 7 Alabama         4.29          335.
## 8 Tennessee       4.28          368.
## 9 Michigan        4.23          307.
## 10 Kentucky       4.06          385.
```

```
us_state_totals %>%
  slice_min(deaths_per_thou, n = 10) %>%
  select(Province_State, deaths_per_thou, cases_per_thou)
```

```
## # A tibble: 10 x 3
##   Province_State      deaths_per_thou cases_per_thou
##   <chr>          <dbl>          <dbl>
## 1 American Samoa    0.611          150.
## 2 Northern Mariana Islands 0.744          248.
## 3 Virgin Islands    1.21           231.
## 4 Hawaii            1.30           269.
## 5 Vermont           1.49           245.
## 6 Puerto Rico        1.55           293.
## 7 Utah              1.65           340.
## 8 Alaska            2.01           415.
## 9 District of Columbia 2.03           252.
## 10 Washington        2.06           253.
```

Death rates vary by roughly a factor of three among the 50 states: Arizona, Oklahoma, and Mississippi all exceeded 4.4 deaths per thousand residents, while Hawaii and Vermont stayed below 1.5. The national average was about 3.4 deaths per thousand.

Modeling the Data

Does knowing a state's case rate let us predict its death rate? We fit a simple linear model of deaths per thousand as a function of cases per thousand across the states and territories.

```
mod <- lm(deaths_per_thou ~ cases_per_thou, data = us_state_totals)
summary(mod)

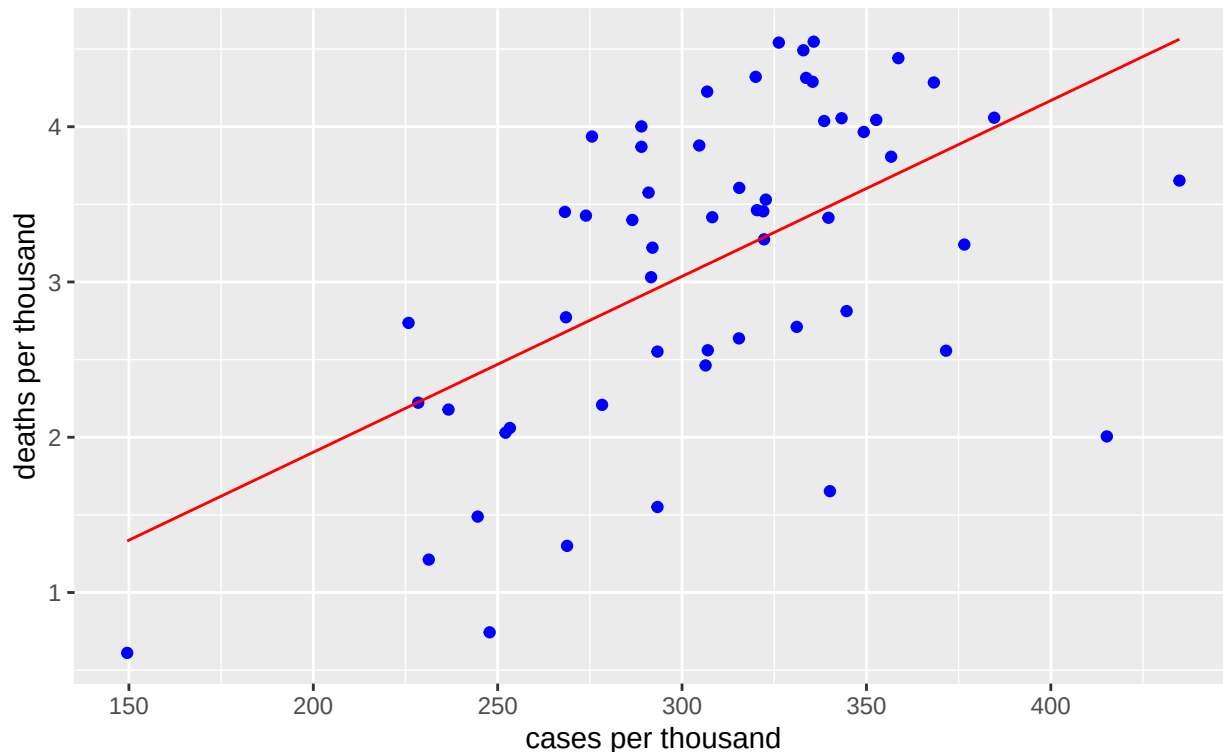
##
## Call:
## lm(formula = deaths_per_thou ~ cases_per_thou, data = us_state_totals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3352 -0.5978  0.1491  0.6535  1.2086
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.36167    0.72480  -0.499    0.62
## cases_per_thou  0.01133    0.00232   4.881 9.76e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8615 on 54 degrees of freedom
## Multiple R-squared:  0.3061, Adjusted R-squared:  0.2933
## F-statistic: 23.82 on 1 and 54 DF,  p-value: 9.763e-06

us_state_totals <- us_state_totals %>%
  mutate(pred = predict(mod))

us_state_totals %>%
  ggplot() +
  geom_point(aes(x = cases_per_thou, y = deaths_per_thou),
    color = "blue") +
  geom_line(aes(x = cases_per_thou, y = pred), color = "red") +
  labs(title = "State Deaths per Thousand vs. Cases per Thousand",
    subtitle = "Blue = actual, red = linear model prediction",
    x = "cases per thousand", y = "deaths per thousand")
```

State Deaths per Thousand vs. Cases per Thousand

Blue = actual, red = linear model prediction



The relationship is positive and statistically significant, but weak: the model explains only about 31% of the variance (R-squared of roughly 0.31). Case rates alone are a poor predictor of death rates. States far above the line (more deaths than their case count predicts) and far below it (fewer) point to other drivers worth investigating: age structure, vaccination rates, hospital capacity, testing intensity, and the timing of when each state was hit — states hit hard in spring 2020 faced the virus before treatments and vaccines existed.

```
us_state_totals %>%  
  mutate(resid = deaths_per_thou - pred) %>%  
  slice_max(abs(resid), n = 6) %>%  
  select(Province_State, cases_per_thou, deaths_per_thou, pred, resid)
```

```
## # A tibble: 6 x 5  
##   Province_State      cases_per_thou deaths_per_thou   pred resid  
##   <chr>             <dbl>         <dbl> <dbl> <dbl>  
## 1 Alaska             415.           2.01  4.34 -2.34  
## 2 Utah               340.           1.65  3.49 -1.84  
## 3 Northern Mariana Islands 248.           0.744 2.45 -1.70  
## 4 Puerto Rico        293.           1.55  2.96 -1.41  
## 5 Hawaii             269.           1.30  2.68 -1.38  
## 6 Guam              372.           2.56  3.85 -1.29
```

Potential Sources of Bias

Several sources of bias should temper the conclusions:

Measurement bias in case counts. Confirmed cases depend on testing. Testing was scarce in early 2020 and changed toward the use of home antigen tests (rarely reported) in 2022, so case undercounting varies over time and by state. This directly affects both the CFR analysis and the regression: some of the state-to-state

variation in “cases per thousand” is variation in testing, not in infection.

Measurement bias in death counts. States differed in how they attributed deaths to COVID-19 (deaths *with* versus *from* COVID), and death certificate practices changed over the pandemic.

Reporting artifacts. The raw data contain day-of-week cycles and occasional bulk corrections (including negative “new case” days when jurisdictions revised totals). The 7-day average mitigates but does not eliminate this.

International comparability. Countries differ enormously in testing capacity, death registration systems, and political incentives to report accurately. Cross-country comparisons of *reported* rates are only meaningful among countries with similar surveillance quality, which is why the global comparison above is framed around wealthy peer countries.

State model The state-level regression cannot be interpreted at the individual level; it also treats each state as one point, ignoring huge within-state variation between counties.

Personal bias. I came to this analysis expecting vaccination-era waves to be less deadly, which is a conclusion the data supports, but I have tried to mitigate confirmation bias by reporting the model’s weak fit honestly and by listing alternative explanations (testing intensity, age structure, timing) rather than attributing state differences to any single cause.

Conclusion

The JHU data show a pandemic that arrived in waves, with the link between infections and deaths weakening dramatically over time: the January 2022 Omicron wave produced the most cases by far, yet daily deaths peaked a full year earlier, and the cumulative CFR fell from about 1.7% to 1.1%. On a per-capita basis the US recorded more deaths than nearly all of its wealthy peers, though cross-country comparisons are limited by reporting differences. Outcomes across states varied threefold on a per-capita basis, and a state’s case rate explains only about a third of the variance in its death rate — meaning most of what determined how deadly the pandemic was in a given state lies in factors beyond raw infection counts. A natural follow-up would be to join in state-level vaccination, age, and hospital-capacity data to model those residuals directly.

I acknowledge the use of AI (Claude Code) to help me check my syntactic constructions in building the code, to search for the correct vocabulary to implementing my intentions where I lacked the choice of right libraries, programming strategy, the need to write a function over the larger dataset, and to help me find the right verb when I needed it. I also used a search engine to help generate ideas for questions to be posed regarding policing work. I relied on the visualisation videos from module 1 to give inspiration to the questions and analysis.

Sample: Thought process Thought process All good questions — let me audit the document against each. The honest answer: six of seven are solidly covered, but the global-vs-US comparison is the weak spot. We import and tidy the global data but only cite world totals in prose — we never actually compare rates. Let me verify numbers for a proper comparison and fix that.

”] Rubric audit, item by item:

Global vs US rates — this was the one gap. We imported global data but only cited totals in prose. Now fixed: a new “How does the US compare globally?” section computes deaths per thousand by country, plots the US against 13 peer countries, and tables the 10 worst (population > 1M). Verified numbers: US ~3.4/thousand, roughly 12th worst of 157 countries — behind Peru (6.7) and Eastern Europe, far worse than Germany (2.0), Canada (1.4), Japan/Korea (<0.7). Stated purpose — yes, “Introduction and Questions of Interest” poses three explicit questions (question 2 now includes the global comparison). Data description — strengthened: the intro now specifies the source, date range, wide format (one row per location, one cumulative-count column per date, county-level for US), and the population lookup table. Model — yes, $\text{lm}(\text{deaths_per_thou} \sim \text{cases_per_thou})$ across states with prediction plot, R^2 discussion, and residual table. Personal bias — yes you have an explicit paragraph on prior expectation and mitigation. Reporting/other biases — yes: testing-dependent case counts, death attribution differences, day-of-week/bulk-correction artifacts, ecological

fallacy, and a new international-comparability paragraph (why we compare only against similar-surveillance peers — e.g., India’s reported 0.4/thousand vs excess-mortality estimates). sessionInfo() — yes, final chunk.”

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Africa/Johannesburg
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0  dplyr_1.2.1
## [5] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2    tibble_3.3.1
## [9] ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0          gtable_0.3.6      crayon_1.5.3      compiler_4.6.1
## [5] tinytex_0.60       tidyselect_1.2.1  parallel_4.6.1    scales_1.4.0
## [9] yaml_2.3.12        fastmap_1.2.0     R6_2.6.1          labeling_0.4.3
## [13] generics_0.1.4     curl_7.1.0        knitr_1.51         pillar_1.11.1
## [17] RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.3.0        utf8_1.2.6
## [21] stringi_1.8.9      xfun_0.60         S7_0.2.2          bit64_4.8.2
## [25] otel_0.2.0         timechange_0.4.0  cli_3.6.6         withr_3.0.3
## [29] magrittr_2.0.5     digest_0.6.39     grid_4.6.1        vroom_1.7.1
## [33] rstudioapi_0.19.0  hms_1.1.4         lifecycle_1.0.5   vctrs_0.7.3
## [37] evaluate_1.0.5     glue_1.8.1        farver_2.1.2      rmarkdown_2.31
## [41] tools_4.6.1        pkgconfig_2.0.3   htmltools_0.5.9
```